

Rohan Praveen Chavan

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PROFESSIONAL SUMMARY

ML engineer specializing in agentic systems, LLM evaluation frameworks, and production AI deployments. Built a production RAG pipeline with LangChain, Llama 3.3 70B, and ChromaDB serving real housing policy researchers - citation grounding score 1.0, 86 automated tests. Fine-tuned LLMs using SFT and DPO on 117K examples, outperforming Claude-3.7 Sonnet on adversarial safety benchmarks (1st place, Amazon Nova Trusted AI Challenge). Shipped LunaFlow, a fullstack agentic assistant with LangChain-orchestrated tool-calling across multiple external APIs, live at lunaflow.work. Deep expertise in the LangChain/LangGraph ecosystem: retrieval pipelines, agent orchestration, inference API selection, and evaluation-driven iteration.

CORE COMPETENCIES

Agentic Systems & LangGraph Orchestration

LLM Evaluation & Monitoring Frameworks

RAG Pipeline Design

Tool-Calling & Multi-Agent Systems

LLM Fine-Tuning (SFT/DPO)

Python & TypeScript (Full-Stack)

Production LLM Deployment

Inference API Optimization

WORK EXPERIENCE

Virginia Center for Housing Research (VCHR), Virginia Tech Apr 2026 - Present

Graduate Research Assistant — Blacksburg, VA

- Engineered production RAG pipeline with **LangChain, Llama 3.3 70B, and ChromaDB** - self-healing fallback (Together AI primary, Groq secondary), two-pass retrieval with 5-tier geographic/income routing, scaling to 6,809 indexed document chunks powering policy decisions for real housing researchers.
- Built **86-test evaluation framework** covering corpus ingestion, retrieval accuracy, and citation grounding - achieved citation grounding score of 1.0 across all validation targets; evaluation system caught retrieval regressions before production deployment across 4 Virginia localities.
- Integrated HUD, BLS, and Census BPS federal APIs with automated startup validation and graceful degradation; identified and replaced unreliable data source, improving pipeline reliability for production deployment.

Virginia Tech (Team HokieTokie) Jan 2025 - Jun 2025

Software Engineer - Amazon Nova Trusted AI Challenge — Blacksburg, VA

- Fine-tuned LLMs using **SFT and DPO on 117K synthetic training examples**, reducing adversarial attack success rate by 46% and outperforming Claude-3.7 Sonnet and CodeLlama-70B on safety benchmarks - 1st place (Tournament 2) among 10 global teams.
- Engineered **adversarial evaluation system** with taxonomy-guided prompt construction and automated quality scoring across vulnerability detection and content filtering; enabled reproducible model experimentation, regression detection, and evaluation-driven iteration at scale.
- Optimized inference latency through model fusion and quantization; deployed dual-expert architecture with sequential filtering and reusable components for production LLM security.

Software Engineering Intern (AI/ML) — San Francisco, CA

- Designed and deployed **production ML scoring microservice** (FastAPI, PostgreSQL, Vertex AI) providing real-time observability across multiple live LLM endpoints in a distributed multi-service architecture serving an 8-person engineering team.
- Automated CI/CD pipeline with pytest suites and JSON Schema validation across agentic workflows; built **reusable prompt engineering components** reducing manual LLM tuning effort by 25%.

Colgate-Palmolive Global Business Services

Jan 2024 - Jun 2024

Process Automation Engineer Intern — Mumbai, India

- Deployed 3 production FastAPI applications serving 400+ employees with 70% automation rate (200+ hours/month saved); reduced response latency from 30s to under 10s via Redis caching and async I/O handling 1K+ daily interactions.

PROJECTS**LunaFlow** Live: lunaflow.work

Shipped fullstack agentic product to real users - React/TypeScript/Node.js/PostgreSQL stack with **LangChain-orchestrated AI assistant** featuring tool-calling across Google Calendar, Google Tasks, and ElevenLabs APIs. Built agent orchestration layer with context window management, explicit tool-calling contracts, automated failure recovery, and OAuth 2.0 authentication end-to-end.

React, TypeScript, Node.js, FastAPI, LangChain, PostgreSQL, Docker, OAuth 2.0

CareRoute Codefest 2025 - 4th/60 teams

Built and deployed **multi-agent healthcare coordination system in 36 hours** - Agent-to-Agent (A2A) protocol for autonomous task delegation across 6 microservices with domain-specific tool invocation, explicit inter-agent contracts, and graceful failure recovery. Deployed to AWS EC2 under competition constraints.

FastAPI, PostgreSQL, Docker, AWS EC2, A2A Protocol, JSON-RPC 2.0

EDUCATION**M.S. Computer Engineering, GPA: 3.71/4.0 — Virginia Tech**

Aug 2024 - May 2026

Coursework: Advanced Machine Learning, Trustworthy ML, Computer Vision, Applications of ML, IoT System Design

B.Tech Information Technology, Distinction, Honors in AI — K.J. Somaiya College of Engineering

Aug 2020 - May 2024

ACHIEVEMENTS

Amazon Nova Trusted AI: Selected as 1 of 10 global teams; 1st place (Tournament 2), 2nd place (Tournament 1)

Codefest 2025: 4th Place + Honorable Mention among 60 teams

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++

AI/ML: LangChain, LangGraph, LLMs, embeddings, RAG pipelines, agentic systems, tool-calling, LLM fine-tuning (SFT, DPO), Hugging Face Transformers, evaluation frameworks

ML Engineering: retrieval, monitoring, model integration, latency/cost optimization, prompt engineering, Llama 3.3 70B, inference API orchestration, experimentation

Backend & Frontend: FastAPI, Node.js, React, RESTful APIs, microservices, WebSockets, CI/CD, Docker

Cloud & Data: AWS (EC2, Lambda, S3, Bedrock), PostgreSQL, ChromaDB, pgvector, Redis, Kubernetes